

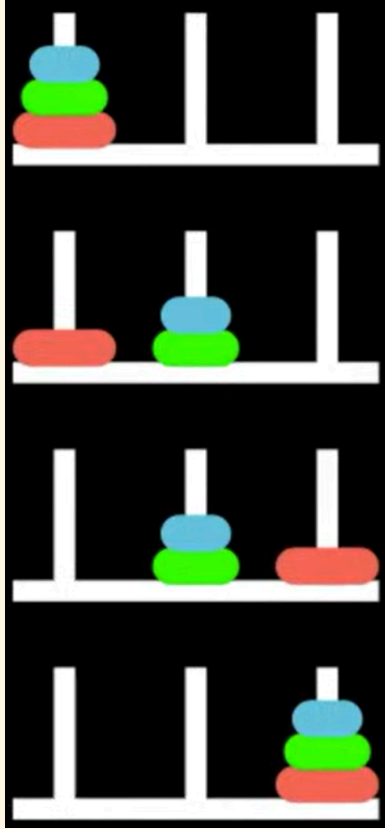
Synthetic Relativity: Tri-modality and the Fixpoint Semantics of the Logical Observer

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The situation calculus

The situation calculus (Towers of Hanoi)



$OnPeg(d, p, s)$

$move(d, p_{from}, p_{to})$

Initial Situation, S_0 :

$OnPeg(D_1, Peg_A, S_0) \wedge$
 $OnPeg(D_2, Peg_A, S_0) \wedge$
 $OnPeg(D_3, Peg_A, S_0)$

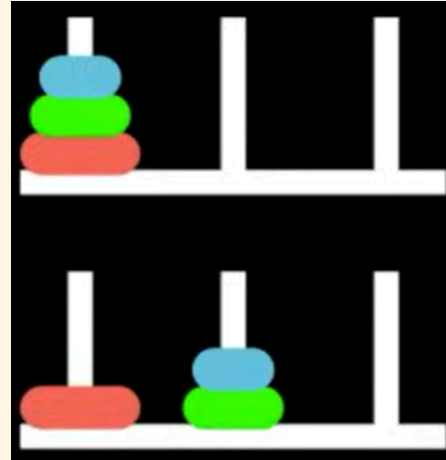
Reiter's Induction Axiom

The Foundational (Second Order) Induction Axiom

$$\forall P. [P(S_0) \wedge \forall a, s (P(s) \rightarrow P(do(a, s)))] \rightarrow \forall s P(s)$$

- Every valid situation is mathematically guaranteed to be a finite sequence of actions rooted at S_0 .
 - Establishes the situation domain as an **inductive** datatype.
-
- Note that standard first order logic cannot enforce reachability due to the Compactness Theorem.

Actions (Situations)



Initial state: S_0

Step 1: $do(move(D_1, Peg_A, Peg_C), S_0)$

Step 2:

$do(move(D_2, Peg_A, Peg_B), do(move(D_1, Peg_A, Peg_C), S_0))$

Step 3:

$do(move(D_1, C, B), do(move(D_2, Peg_A, Peg_B), do(move(D_1, Peg_A, Peg_C), S_0)))$

Situations as a Least Fixpoint

Let $\mathcal{A}$ be actions and $\mathcal{S}$ be the universe of situations. Define a monotonic operator Γ :

$$\Gamma(X) = \{S_0\} \cup \{do(a, s) \mid a \in \mathcal{A}, s \in X\}$$

- $P(S_0) \wedge \forall a, s (P(s) \rightarrow P(do(a, s)))$ translates to $\Gamma(P) \subseteq P$. Here, P is a **pre-fixed point**.
- $\forall s P(s)$ asserts that the universe of situations $\mathcal{S}$ is a subset of P .
- The axiom states $\mathcal{S}$ is a subset of *every* pre-fixed point, meaning it is bounded by their intersection.

Syntactic transformation to First Order Logic

Basic Action Theories [Lin and Reiter 94] define Theory

$\mathcal{D} = \Sigma \cup \mathcal{D}_{S_0} \cup \mathcal{D}_{Poss} \cup \mathcal{D}_{SSA} \cup \mathcal{D}_{UNA}$, where:

- $\mathcal{D}_{S_0}$: axioms describing the initial state of the fluents in S_0
- $\mathcal{D}_{poss}$: axioms describing the prerequisites of the primitive actions, of the

$$Poss(A(\vec{x}), s) \equiv \neg \Theta_A(\vec{x}, s),$$

one for each primitive action A .

- $\mathcal{D}_{ssa}$: axioms describing the effects and non-effects of actions, of the

$$F(\vec{x}, do(a, s)) \equiv \psi_F(\vec{x}, a, s),$$

Action Precondition (Possibility) Axioms

$$\begin{aligned} Poss(move(d, p_{from}, p_{to}), s) \equiv & Disk(d) \wedge (p_{from} \neq p_{to}) \wedge OnPeg(d, p_{from}, s) \\ & \neg \exists d' \left(Disk(d') \wedge Smaller(d', d) \wedge \right. \\ & \left. (OnPeg(d', p_{from}, s) \vee OnPeg(d', p_{to}, s)) \right) \end{aligned}$$

A solution to the frame problem (**sometimes**) [Reiter 91]

1. Positive effect axioms:

$$\exists p_{from} (a = move(d, p_{from}, p)) \supset OnPeg(d, p, do(a, s))$$

2. Negative effect axioms:

$$\exists p_{to} (a = move(d, p, p_{to})) \supset \neg OnPeg(d, p, do(a, s))$$

3. Frame axioms:

  $OnPeg(d, p, s) \wedge \neg \exists p_{to} (a = move(d, p, p_{to})) \supset OnPeg(d, p, do(a, s))$

Successor State Axioms & Regression

So, suppose that for fluent $F(\vec{x}, s)$:

$\gamma_F^+(\vec{x}, a, s)$ encodes all *positive* effects; and

$\gamma_F^-(\vec{x}, a, s)$ encodes all *negative* effects.

The **successor state axiom** solution has the following form:

$$F(\vec{x}, do(a, s)) \equiv \gamma_F^+(\vec{x}, a, s) \vee F(\vec{x}, s) \wedge \neg \gamma_F^-(\vec{x}, a, s)$$

In our example:

$$\gamma_{OnPeg}^+(d, p, a, s) \stackrel{\text{def}}{=} \exists p_{from} (a = move(d, p_{from}, p))$$

$$\gamma_{OnPeg}^-(d, p, a, s) \stackrel{\text{def}}{=} \exists p_{to} (a = move(d, p, p_{to}))$$

Reasoning about actions

- **Projection task:** given a sequence of actions $s = a_1, \dots, a_n$, does con

$$\mathcal{D} \models OnPeg(D_2, Peg_B, do(move(D_2, Peg_A, Peg_B), do(mo$$

- **Deductive planning:** find a sequence of actions to make $\phi(s)$ true (e.g.

$$\mathcal{D} \vdash \exists s. OnPeg(D_1, Peg_C, s) \wedge OnPeg(D_2, Peg_C, s) \wedge OnPe$$

- **Inductive proofs:** proving global invariants across all reachable states
once):

$$\mathcal{D} \models \forall s. executable(s) \supset \forall d, p_1, p_2. (OnPeg(d, p_1, s) \wedge C$$

Property Persistence as a Greatest Fixpoint

Verifying universally quantified temporal queries (e.g., “property ϕ persists indefinitely”) requires second-order induction, however Kelly & Pearce AIJ 2010 show how this can be obtained using a single step regression operator.

The Single-Step Operator: (Kelly & Pearce, AIJ 2010) captures the condition for ϕ holding *now* and being preserved after *any single legal action* via regression ($\mathcal{R}$):

$$\Gamma(Z) \equiv \phi(s) \wedge \forall a (Poss(a, s) \rightarrow \mathcal{R}[Z(do(a, s))])$$

Iteratively applying this meta-level calculation computes the maximal invariant without explicitly invoking the induction axiom.

- By wrapping the single-step regression operator inside a monotonic formula transformer, we evaluate the future back in the current state.
- Taking the greatest fixpoint of this regression operator yields the exact, most permissive preconditions required for the property to persist indefinitely using only first-order logic.

Defining the Observer: Coinduction

Coalgebras formalize the observable system and Coinduction is the proof engine. Let's stop looking at absolute SitCalc coordinates (*OnPeg*) and observe the *relative movement* $\Delta p \equiv (p_{to} - p_{from}) \pmod{3}$ of the smallest disk.

The physical puzzle generates an infinite coalgebraic stream of spatial jumps:

$$\langle +1 \rangle, \langle +1 \rangle, \langle -2 \rangle, \langle +1 \rangle \dots$$

Suppose we want to coinduce an idealized universe where the disk *always* moves $+1$. We mathematically propose the

The Dual Problem & Category Theory

However, to formally synthesize infinite invariants natively, we need a mathematical framework built for continuous forward observation. We must transition from **Algebras** to **Coalgebras**.

In Category Theory, we use an **endofunctor** (F) over a state space (S) as a structural template that shapes our data.

$F(S)$ is simply the resulting combined data object.

- **Algebra (SitCalc / Constructor):** Morphism arrows point

inward: $F(S) \rightarrow S$

- Our endofunctor shapes the input ingredients:

$$F(S) = Action \times S$$

Stone Duality and the Syntactic Transformation

In Category Theory, **Stone Duality** dictates a perfect, inverse mirror-image between Syntax (logical equivalence) and Semantics (topological state space).

When our coalgebra hits the -2 transition, it evaluates the topological future and discovers the physical branching of jumping left twice (-2) is completely indistinguishable from jumping right once ($+1$). We enforce this **Bisimulation Equivalence** by injecting a logical cut:

$$\forall \phi. \langle -2 \rangle \phi \equiv \langle +1 \rangle \phi$$

By *syntactically* equating these propositions, **Stone Duality** automatically forces the underlying *semantic space* to fold

Synthetic relativity axioms

Tri-modality & ternary representation

Standard provability logics like GLB (Gödel-Löb Bimodal Provability Logic) rely on absolute binary provability: xRy assumes an absolute background space where y is provable from x .

- In a ternary structure (drawing from Routley-Meyer semantics for substructural, relevance, and intuitionistic logics), you have a relation $R(x, y, z)$.
- This allows us to decouple the synthetic transformation from the topological space. The ternary relation natively introduces the Observer's frame. It reads: "Relative to reference frame x , the accumulation of inductive evidence y yields the coinductively realised semantic limit z ."

Synthetic Relativity: The Structural Axioms

Dynamic Trimodal Logic with Ternary Relational Semantics:

Axiom 1: The Ternary Relational Axiom Logical entailment is governed by three interacting modalities relative to a specific baseline Reference Frame (S_f):

- **RDo (The Subproblem / Syntax): The Inductive Accumulator ($\Box_1$)** Operates in constructive, non-Boolean logic. Recursively gathers finite, computationally observable properties (compact elements) along a directed logical path. RDo is equivalent to the $\Box$ modality of bimodal provability logic GLB: “*There exists a finite, step-by-step mechanical proof of this property.*”
- **ODo (The Master Problem / Semantics): The Limit/Creator Modality ($\Box_2$)** The semantic evaluator and *interval invariant*. Performs topological closure—evaluating the directed set generated by RDo and snapping it into a discrete, continuous limit. ODo is equivalent to the $\Box_\omega$ modality of GLB: “*Evaluated at the continuous limit, this property is universally true,*” i.e., this is the modality of *semantic reflection*.
- **PDo (Autonomous Progression): The Relativistic Engine ($\Box_3$)** The dynamic transition operator. It orchestrates the recursive interaction of local syntax (RDo) and global semantics (ODo) relative to S_f .

Constructive Accumulation & Realisation

Axiom 2: The Axiom of Constructive Accumulation Within a given reference frame (S_f), the Relativistic Engine (PDo) iteratively drives the state forward, nesting inductive syntax (RDo) and semantic coinductions (ODo):

$$\text{PDo}(S_r, S_o, S_f)$$

$$\text{PDo}(\text{RDo}(R_1, S_r), \text{ODo}(O_1, S_o), S_f)$$

$$\text{PDo}(\text{RDo}(R_2, \text{RDo}(R_1, S_r)), \text{ODo}(O_2, \text{ODo}(O_1, S_o)), S_f)$$

etc.

Theorem I: Semantic Realisation

Because this builds strictly upon finite verifiable observables, autonomous progression strictly halts when the topology reaches Algebraic Compactness. This is the moment of mathematical **Realisation**:

$$\mu(\text{PDo}) = \text{ODo} \left(\bigsqcup_{i \in R} \text{RDo}_i(\perp) \right)$$

- $\bigsqcup \text{RDo}$ (**Syntax**): The builder generates a raw, directed chain of evidence from the ground up. A syntax string alone has no inherent “meaning.”
- ODo (**Semantics**): The top-down evaluator reads the syntax trace and evaluates its truth value.

Proof Sketch: Corresponds to the computation of the **Kleene Fixed-Point Theorem**.

- The ultimate stable fixpoint (μ) is caught exactly when ODo evaluates the limit (Supremum, $\bigsqcup$) of iteratively applying the recursively enumerable inductive proofs (RDo) from baseline ignorance ($\perp$).
- This realised limit acts as the interval semantic invariant.

The Synthetic Transformation

Axiom 3: The Transitivity Axiom Once Realisation is achieved, the relativistic reference frame shifts. The categorically preserved invariant (the semantic limit, O) becomes the new reference frame (S_f) for the next order of progression.

$$\text{PDo}(N, O, S_f) \longrightarrow \text{PDo}(X, Y, O)$$

- **Left Side:** The fully realised state within the initial reference frame (S_f).
- **The Arrow ($\longrightarrow$):** The Synthetic Transformation (Topological closure and Change of Base).
- **Right Side:** A new, higher-order autonomous progression begins. The complex limit O has mathematically synthesised into an atomic, discrete baseline coordinate for the new subproblem (X).

Theorem II: Relativistic Invariance (Preserving the Interval)

Proof Sketch: Category theory mathematically guarantees that under this Transitivity shift (a geometric morphism), the overarching semantic truth-value and geometric integrity of the realised limit (O) are strictly and losslessly preserved across the Change of Base.

The Dual Fixpoints

Unifying Reachability (μ) and Persistence (ν)

By natively intertwining syntax (RDo) and semantics (ODo), the Relativistic Engine mathematically fuses two distinct forms of logical progression:

1. **Induction / Least Fixpoint (μ):** Governed by the RDo modality. The bottom-up synthesis of finite evidence from baseline ignorance ($\perp$). It guarantees computational *Reachability*.
2. **Coinduction / Greatest Fixpoint (ν):** Governed by the ODo modality. The top-down evaluation of continuous semantics. It guarantees *Safety and Property Persistence*.

Algebraic Compactness ($\mu = \nu$) Theorem I (Semantic Realisation) formally computes a **Bilimit**. At the exact coordinate of convergence, the active inductive trace (μ) perfectly fulfills the persistent semantic invariant (ν).

*When this limit shifts to become the new Reference Frame (S_f) via the Transitivity Axiom, **Theorem II** guarantees it acts entirely as a Greatest Fixpoint—a persistent, assumed law of physics for all subsequent computations, natively solving the Frame Problem.*

Synthetic relativity logic

Domain Theory in Logical Form (Syntax versus Semantics)

To execute our modalities, we require an architecture that natively fuses classical (inductive) logic with intuitionistic (coinductive) logic. We ground our system in Abramsky's **DTLF** (1991), based on an **intuitionistic** construction of truth.

- **Baseline Ignorance ($\perp$):** Computation initiates at Bottom ($\perp$)—the mathematical coordinate of zero information.
- **Compact Elements (Finite Observability):** Basic propositions serve as atomic, observable quanta. In DTLF, logical formulas correspond directly to compact elements, meaning any verified property must be confirmable in finite time.
- **Directed Accumulation ($\sqsubseteq$):** Knowledge grows monotonically. The logic is strictly positive and constructive, meaning once an observation is verified, it establishes an upward, directed path toward the limit without retraction.

The Incompleteness Boundary: Logic remains classically monotonic locally. However, due to relative incompleteness (Gödel's Incompleteness & Beklemishev's reflection principles), a static system cannot resolve global limits. The observer must step outside the local system.

Spatial Completeness & The Static Limitation

The mathematical power of vanilla DTLF lies in **Spatial Completeness** (rooted in Stone Duality).

- **The Guarantee:** It proves a 1-to-1 isomorphism between the algebraic, step-by-step **Syntax** ($\phi \vdash \psi$) and the continuous, geometric **Semantics** ($\llbracket \phi \rrbracket \subseteq \llbracket \psi \rrbracket$).
- **Catching Limits:** Computations resolve exactly when the directed path of finite evidence mathematically catches (satisfies) an infinite continuous topological limit (a Supremum).

Vanilla DTLF perfectly describes a single, *static* universe. However, to differentiate induction from coinduction and model dynamic, continuous learning, **the Reference Frame itself must dynamically shift.**

Encoding Planning in Synthetic Relativity (R-DTLF)

Classical planning (e.g., Situation Calculus) relies on a global, static Basic Action Theory (BAT), where rigid evaluation across absolute history leads to combinatorial explosion. In **R-DTLF**, planning is constructed dynamically:

- **Initial Database** $\rightarrow$ **Reference Frame** (S_f): Dictates the local inductive bias—bounding exactly which *fluents* (e.g., **Disk positions in Towers of Hanoi**) are currently observable.
- **The Situation Term** ($do(a, s)$) $\rightarrow$ **Inductive Accumulation** (RDo): Instead of an endlessly nested string of rigid state-jumps, actions are encoded as verifiable proof steps recursively accumulating along a directed path.
- **Goal Formula** ($\exists s. G(s)$) $\rightarrow$ **Semantic Target** (ODo): The goal is not a static Boolean theorem. It is encoded as a **Target Semantic Attractor**—the fluent configuration or overarching invariant the system coinductively attempts to catch.
- **Deductive Search** $\rightarrow$ **Relativistic Engine** (PDo): The autonomous, recursive unrolling of these modalities.

Execution & Invariant Synthesis: Towers of Hanoi example

1. **Local Unrolling (Building the Situation Trace):** The engine recursively accumulates primitive actions (**moving disks**), generating the topological equivalent of a nested $do()$ sequence:

$$\text{PDo}\left(\text{RDo}(\text{move}(D_1, P_2), \dots S_f), \text{ODo}(O_{\text{target}}, \text{Goal}), S_f\right)$$

2. **Limit Catching:** Through top-down semantic reflection ($\square_\omega$), the ODo modality mathematically coinduces a structural property of the optimal traces: **The Parity Invariant (O_{parity})**. *(The smallest disk D_1 must cycle consistently in one fixed direction, while every alternate move is the single legal play between the remaining two pegs, never involving D_1).*

3. **The Synthetic Transformation (Change of Base):** The Transitivity Axiom fires, executing a categorical Change of Base:

$$\text{PDo}(N_{\text{trace}}, O_{\text{parity}}, S_f) \longrightarrow \text{PDo}(X, Y, O_{\text{parity}})$$

Why is the parity invariant found in synthetic relativity (R-DTLF)?

- The alternate move is discovered as the only topologically valid way to accumulate new information.
- In topological terms, reversing the cycle leads to a “dead end” in the domain—a state that cannot act as a supremum for a directed set leading to the goal limit.
- The invariant (cycle D_1 , alternate move, cycle D_1) emerges as the unique, strictly positive proof path through the state space.

In standard classical planners, the Towers of Hanoi puzzle has a continuous branching factor of 3, leading to intractable combinatorial explosion for deep goals.

- The new parity invariant shifts to become the Reference Frame. By adopting this rule as a **lifted axiom**, *the search tree branching factor strictly collapses from 3 down to exactly 1.*
- Note computational complexity of problem doesn't, just *where* the computational effort is spent between a classical recursive approach versus Synthetic Relativity (DTLF) progression.

Translating Continuous Syntax into Successor State Axioms

1. Sidestepping the Frame Problem (SSA vs. R-DTLF)

- **SitCalc SSA (Global Tracking):** Evaluates the entire history (s) across rigid Successor State Axioms:

$$On(d, p, do(a, s)) \equiv a = \text{move}(d, p) \vee (On(d, p, s) \wedge \neg \exists p'. a = \text{move}(d, p'))$$

- **R-DTLF (Local Entailment):** Operates purely on verifiable observables. Irrelevant global variables are topologically invisible (no frame axioms are computed during unrolling):

$$RDo(\text{move}(d, p), S_f) \vdash On(d, p)$$

2. Endogenous Code Generation (Compiling the Precondition) Once O_{parity} is realised, the engine autonomously exports the invariant into a discrete classical **Action Precondition Axiom** ($Poss$) for the new baseline:

$$Poss_{\text{new}}(\text{move}(d, p), s) \equiv Poss_{\text{old}} \wedge (d = D_1 \leftrightarrow \text{LastMoved}(s) \neq D_1)$$

By autonomously synthesising this SSA, the logic enforces deterministic execution.

Bridging Worlds: Endogenous Logic Compilation

R-DTLF does not discard classical logic; it functionally *generates* it at the semantic limits.

Theorem III: Topological Booleanization (Compiling Successor State Axioms)

Upon semantic realisation, the continuous intuitionistic trace generated by RDo can be losslessly compiled into a discrete, classical First-Order baseline via the **Lawvere-Tierney Double Negation ($\neg\neg$) Topology**.

Proof Sketch: Topos theory formally guarantees that inside every fluid, intuitionistic space exists a rigid Boolean sub-lattice that can be safely projected onto.

- **Compiling Successor State Axioms (SSAs):** This geometric compression is mathematically homologous to Ray Reiter's **Explanation Closure**. The Relativistic Engine acts as an endogenous logic compiler—synthesising complex discovered invariants directly into rigid First-Order logical rules.
- **Hybrid Execution:** Because of this formal theorem, we can securely export newly learned R-DTLF heuristics directly into standard classical Basic Action Theories (BATs). This allows R-DTLF to handle complex structural learning, and legacy solvers (like Golog).

Comparison

Synthetic Relativity versus Situation Calculus versus GLB

Aspect	<i>Synthetic Relativity (R-DTLF)</i>	<i>Situation Calculus (BATs/SSAs)</i> + <i>variants of μ-calculus</i>	<i>Bimodal Provability logic (GLB)</i>
Modalities	RDo, ODo & PDo	PDo, $\langle a \rangle \phi$ (possibility), $[a] \phi$ (necessity)	$\Box, \Box_\omega$
Ordering	Sequences of finite Posets, non-unique predecessors, non-well ordered	Successor function, Unique predecessors, non-well ordered	Transfinite ordinality: limits, Non-unique predecessors, well-ordered
Completeness?	Spatially complete (continuous geometry) and Turing complete (universal computation via recursive domains); Arithmetically incomplete (triggers Gödel's incompleteness).	Turing complete (can compute anything), but neither spatially complete (no continuous geometry) nor arithmetically complete (subject to Gödel's incompleteness).	Arithmetically complete (flawlessly models the bounds of provability), but neither Turing complete (strictly decidable, cannot compute loops) nor spatially complete.
Other	Algebraic compactness. Kleene's Realizability & BHK (Brouwer-Heyting-Kolmogorov) Interpretation.	Regression Theorem (solves the frame problem). Fixpoint Semantics & Knaster-Tarski Theorem.	Japaridze's Arithmetical Completeness Theorem. Löb's Theorem & Well-founded Kripke Semantics.